Worksheet # 2 Part B - Upload Solutions
Physics 1

Name: Key Date: _____

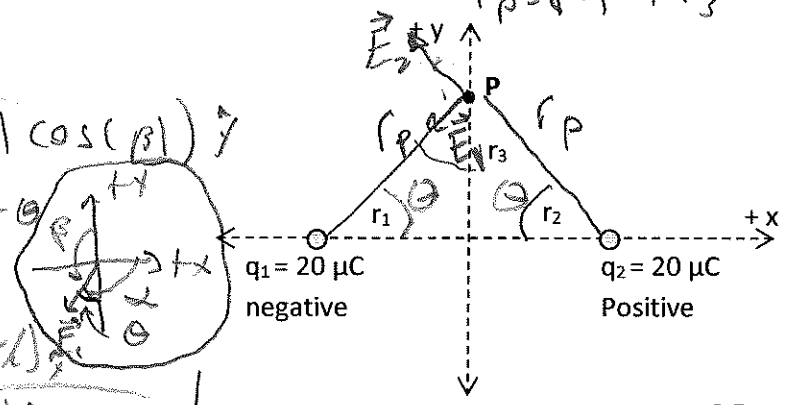
Must show equations used and steps taken in determining the solution for each problem! No work No credit.

Upload the solutions to Questions 11 though 15

11) Using the diagram below, determine the magnitude and direction (α and β) of the net electric field at point P. The net charge is the same for both spheres q = 20 μC but q_1 is a negative charge and q_2 is a positive charge. The distance from the origin to the charge and point P is the same and is $r_1 = r_2 = r_3 = 0.5$ m

$r_p = \sqrt{r_1^2 + r_3^2} = 0.707\,r$

$\Sigma \vec{E}_p = \vec{E}_1 + \vec{E}_2$



$\vec{E}_1 = (|\vec{E}_1|\cos(\alpha))\hat{x} + (|\vec{E}_1|\cos(\beta))\hat{y}$

$\alpha = 90° + \theta \qquad \beta = 180° - \theta$

$\vec{E}_1 = (-254{,}635\ N/C)\hat{x} + (-254{,}635\ N/C)\hat{y}$

$\vec{E}_2 = (|\vec{E}_2|\cos(\alpha))\hat{x} + (|\vec{E}_2|\cos(\beta))\hat{y}$

$\alpha = 90° + \theta \qquad \beta = 90° - \theta$

$\vec{E}_2 = (-254{,}635\ N/C)\hat{x} + (+254{,}635\ N/C)\hat{y}$

$\Sigma \vec{E} = (-509{,}271\ N/C)\hat{x} + (0)\hat{y}$

$|\Sigma \vec{E}| = 509{,}271\ N/C \qquad \alpha = 180° \qquad \beta = 90°$

$r_1 = r_2 = r_3 = 0.5$ m

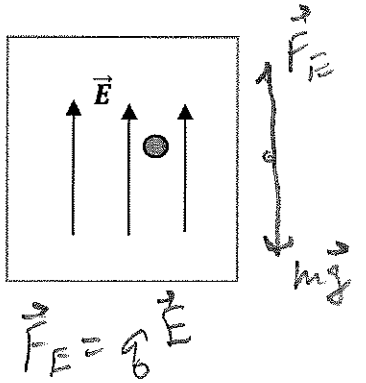
$|\vec{E}_1| = \frac{k q_1}{r_p^2} = 360{,}109\ N/C$

$|\vec{E}_2| = \frac{k q_2}{r_p^2} = 360{,}109\ N/C$

$\theta = \tan^{-1}\left(\frac{r_3}{r_2}\right) = 45°$

12) A ping pong ball (m = 0.005 kg) has a net charge of q, it is placed in a vertical electric field with a strength of 1,000 N/C directed straight up. What net charge q and type of charge (positive or negative) must the ping ball have in order for it to stay stationary in the gravitational field?

q → must be positive for the electric Force to be in the opposite direction as the gravitational Force



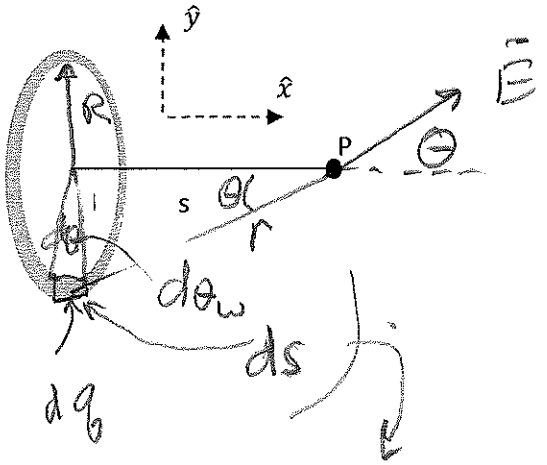
$\Sigma \vec{F} = \vec{F}_E + m\vec{g} = 0$

$\vec{F}_E = q\vec{E}$

$\Sigma F_y = q|\vec{E}| + (-m|\vec{g}|) = 0$

$q = \frac{m|\vec{g}|}{|\vec{E}|} = 4.9 \times 10^{-5}\ C$ positive

13) A circular thin wire with a radius R and a uniform linear charge density of λ is shown below, determine the electric field along the x-axis at point P which is at a distance of s from the center of the wire. Draw and label on the diagram the incremental charge, radius, angle, electric Field and any other variables needed to solve the problem.



$\vec{E} \cdot \hat{x} = |\vec{E}| \cos\Theta$

$dq = \lambda\, dl = \lambda\, ds = \lambda R\, d\theta_w$

$\cos\Theta = \frac{s}{r} = \frac{s}{(s^2+R^2)^{1/2}}$ (constants)

$r = (s^2+R^2)^{1/2}$

$E_x = K\int \frac{\cos\Theta\, dq}{r^2}$

$E_x = K\int_0^{2\pi} \frac{s}{(s^2+R^2)^{1/2}} \cdot \frac{1}{(s^2+R^2)} (\lambda R\, d\theta_w)$

$E_x = \frac{\lambda K s R}{(s^2+R^2)^{3/2}} \left[\theta_w\right]_0^{2\pi} = \frac{2\pi\lambda K s R}{(s^2+R^2)^{3/2}}$

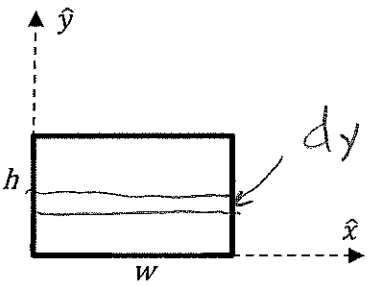
ds - arc length of dq

$ds = R\, d\theta_w$

$d\theta_{win}$ → internal angle from center to dq

14) The electric field magnitude changes along the y-axis by the equation below, determine the equation for the electric flux through the rectangle shown below.

$$\vec{E}(x) = (my^{1/2})\hat{z}$$



$\Phi_E = \int \vec{E} \cdot d\vec{A} = \int_0^h (my^{1/2})(w\, dy)\hat{z}\cdot\hat{z}$

$\Phi_E = mw\int_0^h y^{1/2}\, dy = \frac{mw}{(3/2)}\left[y^{3/2}\right]_0^h$

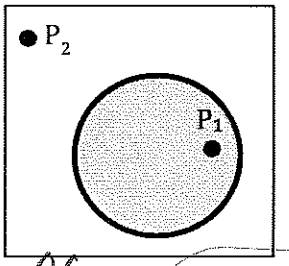
$\Phi_E = \frac{2mw}{3} h^{3/2}$

$d\vec{A} = (w\, dy)\hat{n}$

$d\vec{A} = (w\, dy)\hat{z}$

15) A non-conducting sphere of radius R = 30 cm has a volume charge density of $\rho = 4 \times 10^{-6}$ C/m^3 uniformly distributed throughout its volume.

A) Determine the electric field strength at point P_1, that is at a distance of $r_1 = 20$ cm from the center of the sphere.



$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0} \qquad Q_{in} = \rho V = \rho\left(\tfrac{4}{3}\pi r_1^3\right) \qquad A = 4\pi r_1^2$$

$$|\vec{E}|A = \frac{Q_{in}}{\varepsilon_0} \quad \Big| \quad |\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A} = \frac{\rho\left(\tfrac{4}{3}\pi r_1^3\right)}{\varepsilon_0(4\pi r_1^2)} = \frac{\rho r_1}{3\varepsilon_0} = 3\times10^4 \text{ N/C}$$

B) Determine the electric field strength at point P_2, that is at a distance of $r_2 = 100$ cm from the center of the sphere.

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\varepsilon_0} \qquad Q_{in} = \rho V_s = \rho\left(\tfrac{4}{3}\pi R^3\right) \qquad A = 4\pi r_2^2$$

$$|\vec{E}|A = \frac{Q_{in}}{\varepsilon_0}$$

$$|\vec{E}| = \frac{Q_{in}}{\varepsilon_0 A} = \frac{\rho\left(\tfrac{4}{3}\pi R^3\right)}{\varepsilon_0(4\pi r_2^2)} = \frac{\rho R^3}{3\varepsilon_0 r_2^2}$$

$$|\vec{E}| = 4.07\times10^3 \text{ N/C}$$